Enhanced Surface Soil Moisture Retrieval at High Spatial Resolution From the Integration of Satellite Observations and Soil Pedotransfer Functions

Pei Leng, *Senior Member, IEEE*, Zhao-Liang Li, *Fellow, IEEE*, Qian-Yu Liao, Yun-Jing Geng, Qiu-Yu Yan, Xia Zhang, and Guo-Fei Shang

***Abstract*— Trapezoidal configurations constituted by land surface temperature and fractional vegetation cover have been frequently used to estimate surface soil moisture (SSM). Determination of the SSM status over the trapezoidal dry and wet edges is required to decouple the volumetric SSM content from the trapezoid-derived M_0 because of the coupling of volumetric SSM content and soil texture (i.e., soil moisture availability, M_0). Currently, soil hydraulic characteristics generated from soil pedotransfer functions (PTFs) provide a preferred solution for describing the SSM status over trapezoidal dry and wet edges; however, most PTFs have been developed from laboratory-based soil measurements that have not been fully integrated into remote sensing models for SSM retrieval. This study investigated a practical calibration approach for PTF-derived soil hydraulic characteristics to enhance SSM retrieval using these trapezoidal configurations. Three years of high-resolution SSM measurements were estimated using trapezoidal configurations with Landsat-8 data over a semiarid network in Spain. For the uncalibrated trapezoid, fair accuracy with a root-mean-square error (RMSE) of 0.062 m^3/m^3 and a bias of 0.040 m^3/m^3 was achieved when compared with in situ measurements. Furthermore, a practical PTF-calibration approach with local measurements was proposed and subsequently integrated into the trapezoid to obtain SSM values. Our results indicated enhanced SSM estimates with an RMSE of 0.050 m^3/m^3 and a bias of 0.012 m^3/m^3 with the calibrated PTF. Finally, we found that the calibrated trapezoid can eliminate overestimation and underestimation when the SSM was lower or higher, respectively, which occurred frequently for optical SSM retrievals.**

***Index Terms*— Soil hydraulic characteristics, soil pedotransfer functions (PTF), surface soil moisture (SSM), trapezoid method.**

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Pei Leng, Zhao-Liang Li, Yun-Jing Geng, and Qiu-Yu Yan are with the Institute of Agricultural Resources and Regional Planning, Chinese Academy of Agricultural Sciences, Beijing 100081, China (e-mail: lengpei@caas.cn; lizhaoliang@caas.cn; 82101202277@caas.cn; yanqiuyu@hgu.edu.cn).

Qian-Yu Liao is with the Institute of Geographic Sciences and Natural Resources Research, Chinese Academy of Sciences, Beijing 100101, China (e-mail: qianyuliao@yahoo.com).

Xia Zhang and Guo-Fei Shang are with the School of Land Resources and Urban-Rural Planning, Hebei GEO University, Shijiazhuang 050031, China (e-mail: zhangx396@nenu.edu.cn; shangguofei@hgu.edu.cn).

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I. Introduction

As an essential climate and hydrological variable, surface soil moisture (SSM) plays a significant role in numerous Earth system science applications, such as weather forecasting and water resource management [1], [2]. In particular, knowledge of the spatiotemporal SSM dynamics at high spatial resolution (e.g., tens of meters) is conducive to guiding agricultural activities, such as irrigation schedule and yield prediction [3], [4]. This is especially pivotal for semiarid regions because of the importance of SSM in the determination of vegetation patterns and ecosystem functions in these areas [5], [6], [7].

Currently, two approaches are widely used to obtain SSM at high spatial resolution. The first is a direct method that utilizes quantitative retrieval algorithms using high-spatial-resolution optical images or synthetic aperture radar (SAR) observations. The other is an indirect method relying on disaggregation algorithms with coarse-spatial-resolution SSM products [8]. Recently, many efforts have been dedicated to producing global SSM products at coarse spatial resolutions from microwave measurements, allowing for an unprecedented opportunity to obtain high-spatial-resolution SSM data using disaggregation algorithms [9], [10]. Nevertheless, most currently available disaggregation algorithms still require optical or SAR observations as auxiliary inputs; therefore, direct high-spatial-resolution SSM estimates via quantitative retrieval algorithms are encouraged in practice since complicated disaggregation algorithms may introduce more uncertainties due to the fusion of signals from different wavelengths and neglect of scaling issues [11]. In recent years, several studies have highlighted the feasibility of producing relatively high spatial resolution (~1–3 km) SSM products based on the Sentinel-1 observations. A typical dataset is the 3-km Level-2 (L2) SSM product from the synergetic use of the soil moisture active passive (SMAP) radiometer and the Sentinel-1A/Sentinel-1B radar (known as SMAP/Sentinel-1 SSM product) [12]. Another kilometer-scale SSM product is the one based on the Sentinel-1 and the Advanced Scatterometer observations, which is now available in the Copernicus Global Land Monitoring Services (https://land.copernicus.eu/global/) [13], [14]. However, all these products are characterized by a spatial resolution of a few kilometers, which is not directly applicable at the field scale.

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Compared to SAR-based SSM estimations, optical-data-based SSM retrievals have received more attention in practice, primarily because of the ease of data acquisition and processing and the well-developed methodology involving quantitative retrieval algorithms. Among the currently available optical SSM retrieval algorithms, trapezoid-based methods have recently become a frequently used approach for deriving SSM at different spatial scales [15], [16]. Although several studies have highlighted the acceptable accuracies of trapezoid-based SSM retrieval [17], [18], it is widely recognized that SSM is an indirect variable that should be extracted from the direct output of satellite-derived SSM proxies. The direct trapezoidal output is the soil moisture availability (M_0), which is interpolated between the trapezoidal dry edge with $M_0 = 0$, that is, zero water availability for evapotranspiration, and trapezoidal wet edge with $M_0 = 1$, that is, plenty of free water for evapotranspiration. Hence, a critical issue in trapezoid-based SSM retrieval is determining the accurate SSM status over dry and wet edges.

Previous studies have confirmed that SSM generally varies between two critical soil hydraulic characteristics under normal circumstances, that is, the wilting point and field capacity [19], [20], [21]. This is adequate in crop or soil models where volumetric SSM contents at the wilting point and field capacity can infer the lower and drained upper limits of water status, respectively [22], [23]. Consequently, these two soil hydraulic characteristics have been widely assumed to approximate the volumetric SSM content over trapezoidal dry and wet edges. In theory, accurate soil hydraulic characteristics should be measured under field and laboratory conditions; however, field and laboratory measurements of soil hydraulic characteristics are often difficult and impractical for agricultural hydrology or water management in natural landscapes. Other approaches for approximating these hydraulic characteristics through ground in situ state variables, such as SSM dynamics and pressure heads over the long term, would also confront these difficulties [24], [25]. Bouma [26] first introduced pedotransfer functions (PTFs) to estimate soil hydraulic characteristics from easily measured soil attributes, such as texture fractions, organic matter, bulk density, and soil structure. Several PTFs have since been developed to obtain soil hydraulic characteristics, most of which are based on soil surveys for specific regions [27], [28]. Consequently, direct extrapolation of these regional PTF-based soil hydraulic characteristics to other regions may result in errors. In a previous study based on a crop model simulation and a set of field-measured data, it was summarized that the approach proposed by Saxton et al. [29] performed the best among eight widely used PTFs for calculating soil water retention characteristics [22]. Nevertheless, many soil sciences and hydrological applications are still concerned with the performance of PTF-derived soil water retention characteristics and the calibration of these characteristics to improve the simulation of SSM variations and related agricultural hydrology activities, such as irrigation scheduling and drought monitoring over semiarid regions [30], [31], [32], [33].

Similar to agricultural hydrology applications, remote-sensing-based algorithms also require the calibration of soil hydraulic characteristics to improve SSM retrieval since satellite-derived SSM proxies are directly coupled with soil water retention and spatial extrapolation of local PTF-derived soil hydraulic characteristics into satellite pixel scales is in high demand. In particular, numerous digital soil maps have been produced at various spatial scales, allowing for an unprecedented opportunity to derive volumetric SSM content using remote-sensing-based SSM retrieval algorithms; however, this issue has not been adequately investigated, especially for the widely used trapezoid-based SSM retrieval methods over semiarid regions, where accurate SSM estimates have unique importance for the areas. Moreover, although most field SSM measurements do not exceed the field capacity, it is still unclear which SSM status should be accounted for over the trapezoidal wet edge since SSM can remain a level above the field capacity at times, such as after irrigation or heavy rain. This study aimed to address these issues with synchronous field measurements and quasi-matched high-spatial-resolution satellite observations to provide adequate solutions for SSM retrieval improvements.

Overall, this study makes assessments of high-spatial-resolution SSM retrievals with in situ measurements to determine the preferred approximation scheme for SSM status determination over trapezoidal boundaries for a semiarid region. Furthermore, this study develops a practical PTF-calibration approach to improve the estimation of soil hydraulic characteristics and integrate them into the trapezoid method to enhance SSM retrieval. Section II presents the materials and methods. The results and discussion are presented in Sections III and IV, respectively, and the conclusion is provided in Section V.

II. Materials and Methods

A. Study Region and in Situ Measurements

In this study, a well-organized SSM network named REd de MEDición de la HUmedad del Suelo (REMEDHUS) in the semiarid region of Spain was selected to investigate trapezoid-based SSM retrieval. Located in the central part of the Duero Basin, the REMEDHUS network covers a relatively flat area (slope less than 10°) of approximately 1300 km^2. The REMEDHUS region has a continental semiarid Mediterranean climate with a mean air temperature of 12 °C and an average annual precipitation of 385 mm [34]. The REMEDHUS network is a member of the International Soil Moisture Network (ISMN) and has been selected as a core calibration/validation site for many microwave-based SSM products. In the past decade, several studies have reported the use of in situ SSM measurements collected at the REMEDHUS network for the validation of satellite-derived SSM datasets [35], [34]. Fig. 1 shows the distribution of the in situ SSM sites of the REMEDHUS network. In this study, ground-based in situ SSM measurements with a temporal resolution of 1 h at 20 stations during the growing season from April to October each year between 2013 and 2015 were collected to evaluate the satellite-derived SSM. Table I lists the SSM sites in the REMEDHUS network. For each SSM site, soil texture information was measured by laboratory analyses of soil samples collected around

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TABLE I
BRIEF INFORMATION OF THE SSM SITES AT THE REMEDHUS NETWORK

Site ID	Latitude	Longitude	Altitude (m)	Sand(%)	Silt(%)	Clay(%)
E10	41.27611°N	5.59056°W	870	75.11	16.35	8.54
F06	41.37455°N	5.54714°W	745	67.19	13.70	19.11
F11	41.2404°N	5.54291°W	855	81.52	11.97	6.51
H07	41.35004°N	5.48891°W	755	85.10	9.64	5.26
H09	41.2905°N	5.43402°W	720	19.78	44.99	35.23
H13	41.18381°N	5.47572°W	900	70.36	11.45	18.19
I06	41.38251°N	5.42786°W	720	89.81	5.93	4.26
J03	41.45703°N	5.40964°W	665	85.05	11.26	3.69
J12	41.20843°N	5.41224°W	835	60.94	16.85	22.21
J14	41.15011°N	5.39621°W	830	66.81	20.98	12.21
K04	41.42529°N	5.37267°W	740	87.09	9.27	3.64
K09	41.3069°N	5.35925°W	720	74.36	15.00	10.64
K10	41.26611°N	5.37972°W	745	91.16	5.71	3.13
K13	41.1972°N	5.35861°W	720	62.22	18.21	19.57
L03	41.44765°N	5.35734°W	675	82.25	6.44	11.31
L07	41.35873°N	5.32977°W	790	46.80	20.78	32.42
M05	41.39508°N	5.3201°W	750	81.64	8.31	10.05
M09	41.28662°N	5.29868°W	795	49.83	24.89	25.28
N09	41.30126°N	5.24569°W	765	62.46	16.78	20.76
O07	41.34778°N	5.22361°W	770	78.84	13.47	7.69

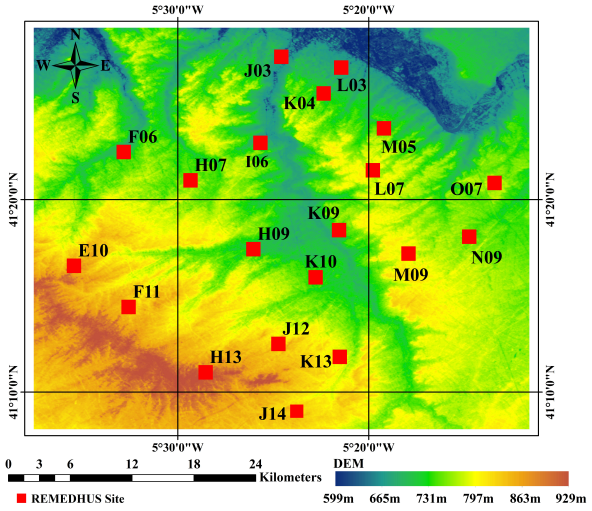


Fig. 1. Distribution of SSM sites over the REMEDHUS network.

the site. All REMEDHUS data in this study were obtained from the ISMN website (https://ismn.geo.tuwien.ac.at/en/).

B. Satellite Data

The Landsat spacecraft series from the U.S. National Aeronautics and Space Administration (NASA) has the longest continuous Earth imaging functionality, that is, nearly 50 years, starting with the launch of Landsat-1 in 1972. The latest Landsat mission is Landsat-8, which was launched in February 2013. The Landsat data archive continues expanding with Landsat-9 scheduled for launch in 2023. Specifically, the Landsat-8 Operational Land Imager/Thermal Infrared (OLI/TIRS) products have been available since April 2013, which matches the study period of this investigation. Readers can refer to Cook et al. [36] and Malakar et al. [37] for the detailed description of the algorithm for retrieving Landsat-8 land surface temperature (LST). To decide the practicable satellite data, we made a simple rule that the Landsat-8 image for each selected date should include at least five valid Landsat-8 pixels (corresponding to five REMEDHUS sites) for SSM retrieval. During this period, 40 Landsast-8 images with operational LST and surface reflectance products at a high spatial resolution of 30 m were collected from the official website of the U.S. Geological Survey (https://www.usgs.gov/Landsat-missions). Specifically, the normalized difference vegetation index (NDVI) was primarily calculated from the surface reflectance product of Landsat-8 at bands 4 and 5. Furthermore, the Landsat-8 NDVI was used to obtain the fractional vegetation cover (FVC) following the method proposed by Prihodko and Goward [38]. Fig. 2 shows the day of year (DOY) of the Landsat-8 data collected in this study.

C. Determination of SSM From Trapezoid Configurations

Trapezoid-based methods have been widely recognized as one of the main optical approaches to SSM retrieval. The physical basis of the trapezoidal configuration constituted by scatterplots of spatially distributed LST and FVC can be explained by the decrease in sensitivity of SSM content to LST with an increase in FVC [15]. Fig. 3 shows a schematic of the trapezoid envelope based on satellite-derived LST and FVC data within a pixel window. For the trapezoid-based SSM retrieval method, a dry and wet edge should be confirmed first. In this study, the algorithm proposed by Tang et al. [39] was used to determine these dry and wet edges. Following the algorithm, these edges can be expressed as functions of

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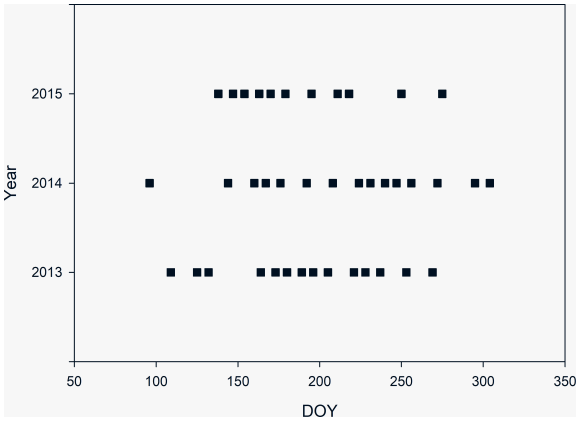


Fig. 2. Landsat-8 data for each DOY used in this study.

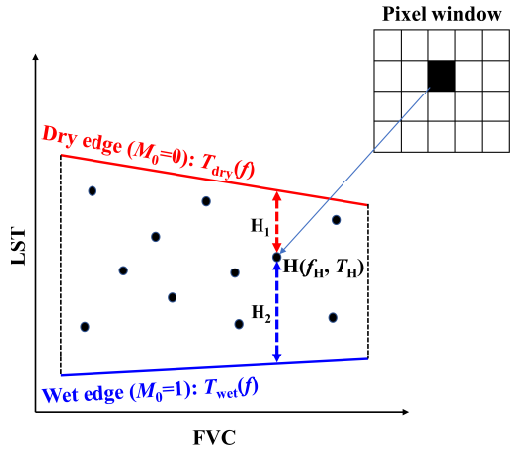


Fig. 3. Summary of the key descriptors and physical interpretations of the LST–FVC trapezoid. M_0 refers to soil moisture availability. M_0 over the dry edge and wet edge response to zero water availability for evapotranspiration and plenty of freedom water for evapotranspiration in theory.

LST–FVC as follows:

$$T_{\text{dry}}(f) = a_1 \times f + b_1 \tag{1}$$

$$T_{\text{wet}}(f) = a_2 \times f + b_2 \tag{2}$$

where $T_{\text{dry}}(f)$ and $T_{\text{wet}}(f)$ are the functions of the LST on the dry and wet edges to f, which represents the FVC variable, respectively; a_1 and a_2 are the slopes of the linear functions of the dry and wet edges, respectively; and b_1 and b_2 are the intercepts of the linear functions of the dry and wet edges, respectively.

For any given pixel (e.g., pixel H with an LST of T_H and an FVC of f_H) in the trapezoid, the connecting lines of the pixel to the dry and wet edges are defined as follows:

$$H_1 = (a_1 \times f_H + b_1) - T_H \tag{3}$$

$$H_2 = T_H - (a_2 \times f_H + b_2) \tag{4}$$

where H_1 and H_2 are the lines connecting pixel H to the dry and wet edges, respectively.

As a result, M_0 of pixel H can be interpolated with the dry edge with $M_0 = 0$ and the wet edge with $M_0 = 1$ as follows:

$$M_0 = \frac{H_1}{H_1 + H_2}. \tag{5}$$

After calculating M_0 for the target pixel using satellite-derived LST and FVC values, M_0 needed to be converted to volumetric SSM content. M_0 is assumed to vary linearly between the dry and wet edges, allowing the volumetric SSM content for the target pixel, which can be written as follows:

$$\theta = M_0 \times \left(\theta_{\text{wet}} - \theta_{\text{dry}}\right) + \theta_{\text{dry}} \tag{6}$$

where θ is the volumetric SSM content and θ_{dry} and θ_{wet} are the volumetric SSM contents over the dry and wet edges, respectively.

D. Integration of Trapezoid Dry and Wet Edges With PTF

As shown in Fig. 3, M_0 can be derived from a trapezoid composed of satellite images. Consequently, θ_{dry} and θ_{wet} are the only parameters that need to be determined before applying trapezoid-based methods in practice. Following the trapezoidal configuration, the dry edge represents $M_0 = 0$, whereas the wet edge represents plenty of free water for evapotranspiration. Nevertheless, since the residual SSM content (θ_r) and saturated SSM content (θ_s) are hydraulic parameters that are primarily determined under laboratory conditions, extremely low (i.e., residual) or high (i.e., saturated) values are most unlikely to occur in field or remotely sensed observations. In contrast, volumetric SSM content generally varies between the wilting point (θ_{wp}) and field capacity (θ_{fc}) under normal circumstances. In this study, two schemes for SSM status approximations over dry and wet edges were investigated: 1) $\theta_{\text{dry}} = \theta_{\text{wp}}$ and $\theta_{\text{wet}} = \theta_{\text{fc}}$ and 2) $\theta_{\text{dry}} = \theta_{\text{wp}}$ and $\theta_{\text{wet}} = (\theta_{\text{fc}} + \theta_{\text{s}})/2$. The second scheme is generally justified because field SSM measurements may exceed field capacity for a while in some cases, such as after irrigation or heavy rain.

In this study, the detailed soil texture information, including the sand and clay percentages, was collected for each REMEDHUS site (Table I); therefore, the PTF proposed by Saxton et al. [29] was used to estimate the three soil hydraulic parameters (θ_{wp}, θ_{fc}, and θ_s). The Saxton et al. [29] method was selected because previous literature showed that it performed best among eight PTFs at the time for estimating water retention parameters with loamy sand, sandy loam, loam, and silt loam soils [22]. Fig. 4 shows the three soil hydraulic characteristics (θ_{wp}, θ_{fc}, and θ_s) obtained following the method of Saxton et al. [29] for the REMEDHUS sites.

E. Calibration of PTF-Derived Soil Hydraulic Characteristics

Because PTF-derived soil hydraulic characteristics may reveal various discrepancies with actual values, the calibration of these characteristics to improve agricultural and hydrological applications is commonly necessary [30], [31], [32], [33].

In the present study, the ground measurements of site-specific soil texture and long-term in situ SSM provide a unique opportunity to calibrate the PTF-derived soil hydraulic characteristics (θ_{wp} and θ_{fc}) for improving trapezoid-based SSM retrievals. Specifically, long-term ground-based in situ SSM measurements from the initial observation (primarily from 2005) to 2021 were collected to determine the maximum and minimum SSM values for each REMEDHUS site. For

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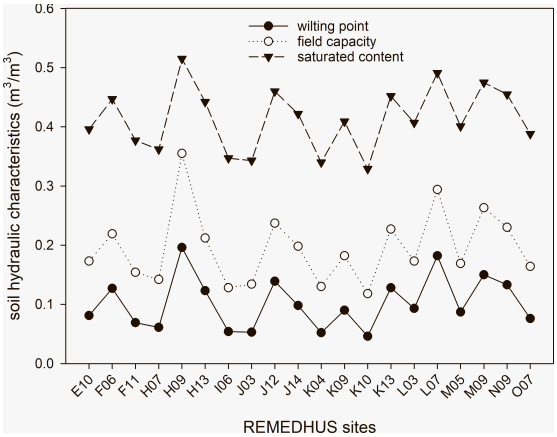


Fig. 4. Three soil hydraulic characteristics (wilting point, field capacity, and saturated content) estimated from in situ soil texture measurements for the REMEDHUS sites following the PTF by Saxton et al. [29].

the long-term ground-based in situ SSM measurements for each REMEDHUS site, a minimum percentile level of 5% and a maximum percentile level of 95% were used to filter SSM outliers. The rationale behind this lies in the expectation that extreme quantiles better reflect theoretical dry and wet SSM statuses as they approach the distribution extremes. SSM normally varies between the wilting point and field capacity; therefore, the in situ SSM measurements at the percentile levels of 5% and 95% can be reasonable to approximate θ_{dry} and θ_{wet} in this study, respectively.

Fig. 5 shows the scatter of the PTF-derived soil hydraulic characteristics by Saxton et al. [29] versus those obtained from our long-term ground-based in situ SSM measurements for the REMEDHU sites. According to the results, it is obvious that the PTF-derived θ_{wp} and θ_{fc} correlate with the long-term ground-based in situ SSM measurement-derived θ_{dry} and θ_{wet}, respectively, with considerable correlation coefficients of 0.79 and 0.89. Nevertheless, a significant overestimation with a bias of 0.058 m^3/m^3 for the wilting point and an overall underestimation with a bias of -0.033 m^3/m^3 for the field capacity were found. Notably, most of the REMEDHUS sites revealed underestimation with regard to the PTF-derived field capacity, especially for sites with higher field capacity. In addition, a critical value of approximately 0.18 m^3/m^3 can also be found to divide the PTF-derived field capacity into sections of overestimation and underestimation when compared to those obtained from long-term ground-based in situ SSM measurements.

As the PTF-derived soil hydraulic characteristics correlated well with those obtained from long-term in situ ground measurements, the PTF-derived soil hydraulic characteristics were calibrated to agree more closely with the ground-measured values as follows:

$$\overset{\Lambda}{P}_i = m \times P_i + n \tag{7}$$

where $\overset{\Lambda}{P}_i$ and P_i refer to the calibrated soil hydraulic characteristics (θ_{wp} and θ_{fc}) and the original PTF-derived soil hydraulic characteristics for the ith site in the REMEDHUS network,

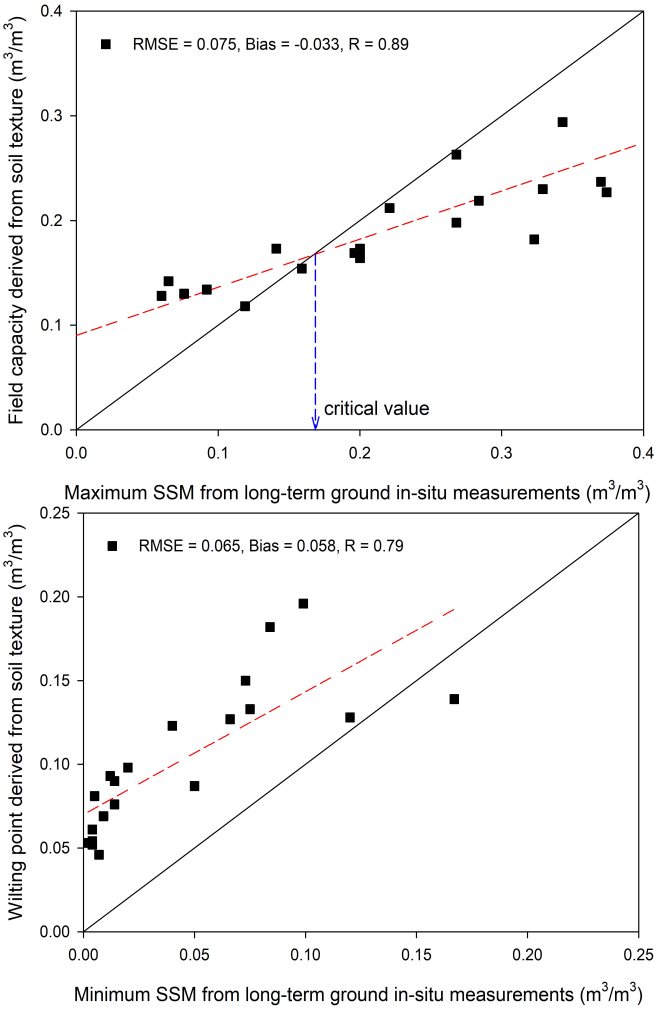


Fig. 5. Comparison of the PTF-derived soil hydraulic characteristics (θ_{wp} and θ_{fc}) and those obtained from long-term (2005–2021) ground in situ measurements for the REMEDHUS sites.

respectively, and m and n are the calibration coefficients that are set such that the RMSE between the estimated and measured values is minimized.

III. Results

A. SSM Retrieval With Uncalibrated Trapezoid

Following the trapezoid-based SSM retrieval method, all valid Landsat-8 pixels within the REMEDHUS network were first combined to constitute the trapezoidal pixel window for each investigated day during the study period. Furthermore, for the Landsat-8 pixels where the REMEDHUS sites reside, M_0 was derived based on (5). Specifically, the relatively high spatial resolution (30 m) of the Landsat-8 images makes it reasonable to assume that the soil hydraulic parameters (θ_{wp}, θ_{fc}, and θ_s) obtained at the site scale are applicable to the Landsat-8 pixels. Finally, the volumetric SSM content was estimated using a trapezoid-based SSM retrieval method. Fig. 6 shows the scatterplots between the in situ SSM measurements and the estimated values, considering the two approximation schemes of the SSM status over the dry and wet edges.

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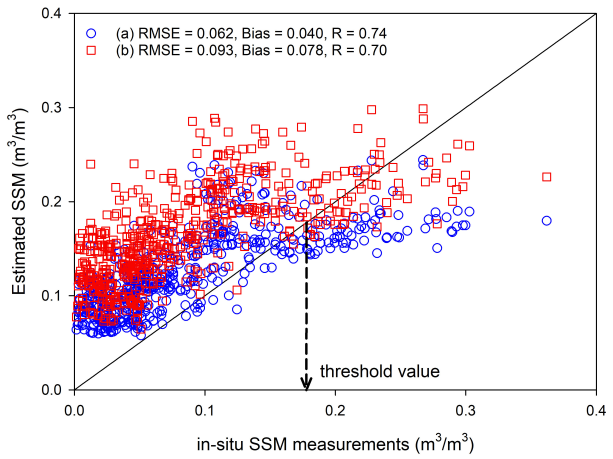


Fig. 6. Comparison of estimated SSM and in situ measurements. (a) and (b) Different approximations of the SSM status over the dry and wet edges. For (a), $\theta_{dry} = \theta_{wp}$ and $\theta_{wet} = \theta_{fc}$. For (b) $\theta_{dry} = \theta_{wp}$ and $\theta_{wet} = (\theta_{fc} + \theta_s)/2$. θ_{wp}, θ_{fc}, and θ_s were obtained with soil texture collected at the REMEDHUS sites following the method proposed by Saxton et al. [29].

TABLE II

ACCURACY STATISTICS (UNIT: m^3/m^3) FOR THE TRAPEZOID-BASED SSM RETRIEVAL WITH TWO APPROXIMATION SCHEMES FOR THE SSM STATUS OVER THE DRY AND WET EDGES. (A) $\theta_{dry} = \theta_{wp}$ AND $\theta_{wet} = \theta_{fc}$. (B) $\theta_{dry} = \theta_{wp}$ AND $\theta_{wet} = (\theta_{fc} + \theta_s)/2$. THE THRESHOLD VALUE IS PROVISIONALLY DETERMINED AS 0.18 m^3/m^3 FOLLOWING THE RESULTS SHOWN IN FIG. 5

	scheme (a)		scheme (b)	
	RMSE	**Bias**	**RMSE**	**Bias**
$\theta > 0.18$	0.076	-0.065	0.045	-0.021
$\theta < 0.18$	0.061	0.051	0.096	0.088

Although the two approximation schemes for the dry and wet edges and SSM conditions revealed similar correlation coefficients (R) of 0.74 and 0.70, respectively, scheme (a) obtained overall better SSM estimates with a root-mean-square error (RMSE) of 0.062 m^3/m^3 and a bias of 0.040 m^3/m^3 and scheme (b) exhibited a decreased accuracy with an RMSE of 0.093 m^3/m^3 and a bias of 0.078 m^3/m^3. Overall, both approximation schemes showed considerable overestimation for SSM retrieval over the REMEDHUS region. Notably, when SSM exceeded a provisional threshold value (approximately 0.18 m^3/m^3), scheme (a) significantly underestimated SSM, whereas scheme (b) corrected the underestimation and obtained better SSM estimates. Meanwhile, although both approximation schemes overestimated SSM when SSM was lower than the provisional threshold value, the SSM estimated using scheme (a) was more accurate than that using scheme (b). Table II summarizes the accuracy statistics for the trapezoid-based SSM retrieval with two SSM status approximation schemes over the dry and wet edges, with an emphasis on accuracy, aside from the provisional threshold value of approximately 0.18 m^3/m^3. Scheme (b) significantly decreased the bias from -0.065 m^3/m^3 with scheme (a) to -0.021 m^3/m^3 when the SSM exceeded the provisional threshold value, indicating that the approximation of SSM status over the wet edge with a value greater than the field capacity was generally reasonable under certain conditions, such as after irrigation or heavy rain. Notably, since the REMEDHUS region is dominated by a semiarid climate pattern, a high percentage (>90%) of the ground in situ SSM measurements would not exceed the provisional threshold value of approximately 0.18 m^3/m^3. Hence, in most cases, a fair accuracy with an RMSE of approximately 0.06 m^3/m^3 and a bias of approximately 0.05 m^3/m^3 was achieved for the trapezoid-based SSM retrieval with consideration of the SSM varying between the wilting point and the field capacity.

B. SSM Retrieval With Calibrated Trapezoid

Section III-A estimated a provisional threshold value of approximately 0.18 m^3/m^3 according to Fig. 6. The basis for determining the provisional threshold value is the conspicuously observed phenomenon in which the threshold value generally divides SSM retrievals into overestimation and underestimation when the SSM is lower or higher, respectively. This phenomenon has also been observed in many optical-based SSM retrieval methods [40], [41]. With the determination of the provisional threshold value in the previous section lacking further support, it is necessary to investigate this issue in more depth.

Because the critical value divided the PTF-derived field capacity into sections of overestimation and underestimation when compared to those obtained from long-term ground-based in situ SSM measurements as shown in Fig. 5, to further investigate the phenomenon of overestimation and underestimation when SSM is lower or higher, respectively, (6) was transformed as follows:

$$\theta = M_0 \times \theta_{wet} + (1 - M_0) \times \theta_{dry}. \tag{8}$$

Based on Fig. 5, for REMEDHUS sites with field capacities lower than the critical value, both θ_{dry} and θ_{wet} revealed significant overestimation, which would lead to the overestimation of SSM directly, according to (6). However, once the field capacity exceeded the critical value, an underestimation of SSM appeared, which was mostly attributed to the underestimation of field capacity because of its dominant role in the determination of SSM in (8).

The critical value in Fig. 5 is close to the provisional threshold value in the previous section, making it reasonable to assume that the PTF-derived soil hydraulic characteristics have significant effects on the trapezoid-based SSM retrieval. As such, PTF-derived soil hydraulic parameters may need to be calibrated using local measurements in practice. To this end, the method for calibrating the PTF-derived soil hydraulic characteristics presented in the previous section has been implemented for all the REMEDHUS sites. Table III shows the calibration coefficients for the Saxton et al. [29] method for estimating the volumetric SSM content at the wilting point and field capacity. Specifically, the long-term ground-based in situ SSM measurements at percentile levels of 5% and 95% were regarded as the measured θ_{wp} and θ_{fc}, respectively.

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TABLE III

CALIBRATION COEFFICIENTS FOR THE PTF-DERIVED SOIL HYDRAULIC CHARACTERISTICS BY THE SAXTON et al. [29] METHOD OVER THE REMEDHUS SITES

Soil hydraulic characteristics	*m*	*n*
Wilting point	0.82	-0.04
Field capacity	1.73	-0.11

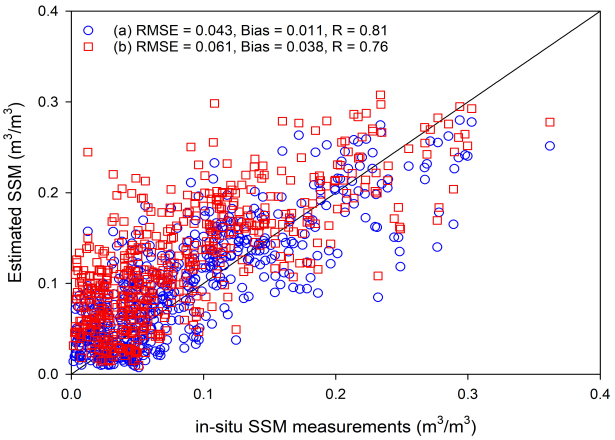


Fig. 7. Comparison of estimated SSM and in situ measurements. (a) and (b) Different approximations of the SSM status over the dry and wet edges. For (a), $\theta_{dry} = \theta_{wp}$ and $\theta_{wet} = \theta_{fc}$. For (b) $\theta_{dry} = \theta_{wp}$ and $\theta_{wet} = (\theta_{fc} + \theta_s)/2$. θ_{wp}, θ_{fc}, and θ_s are determined as the long-term ground-based in situ SSM measurements at percentile levels of 5% and 95%, respectively.

Fig. 7 shows the scatterplots between the in situ SSM measurements and the estimated values with the consideration of the two SSM content approximation schemes over the dry and wet edges, with a special emphasis that θ_{fc} and θ_s are determined as the long-term ground-based in situ SSM at percentile levels of 5% and 95%, respectively. It is evident that improved SSM estimates can be obtained with both schemes compared to those presented in Fig. 6. Specifically, a high accuracy with an RMSE of 0.043 m^3/m^3 and a negligible bias of 0.011 m^3/m^3 can be obtained for scheme (a), which further confirms that SSM generally varies between the wilting point and field capacity [19], [20]. For scheme (b), although it is assumed that there is a wider range of possible SSM dynamics, considerable accuracy with an RMSE of 0.061 m^3/m^3 and a bias of 0.038 m^3/m^3 can also be found. Notably, the accuracy statistics for scheme (b) in this situation (RMSE = 0.061 m^3/m^3, bias = 0.038 m^3/m^3, and $R = 0.76$) are even better than those for scheme (a) in cases where soil hydraulic characteristics are derived from uncalibrated PTF, as shown in Fig. 5 (RMSE = 0.062 m^3/m^3, bias = 0.040 m^3/m^3, and $R = 0.74$). These results indicate that, for trapezoid-based SSM retrieval, the accuracy of the PTF-derived soil hydraulic characteristics may be more affected than the approximation schemes for the SSM status over the dry and wet edges. Furthermore, overestimation and underestimation when SSM is lower or higher, respectively, than the provisional threshold value have been eliminated, which further confirms the necessity for PTF calibration. Moreover, this phenomenon is most likely caused by inadequate PTF for describing the soil water retention characteristics.

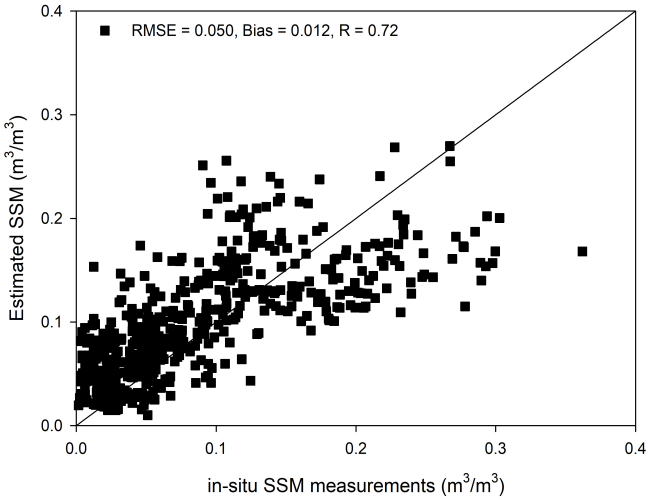


Fig. 8. Comparison of estimated SSM and in situ measurements. The PTF-derived soil hydraulic characteristics by Saxton et al. [29] were calibrated with in situ SSM measurements (2005–2021).

Based on the calibrated θ_{wp} and θ_{fc} for the REMEDHUS sites, the volumetric SSM contents were further estimated and compared with in situ measurements (Fig. 8). The calibrated method, compared to the uncalibrated method with an RMSE of 0.062 m^3/m^3 and bias of 0.040 m^3/m^3 shown in Fig. 5, had a significantly lower RMSE (0.050 m^3/m^3) and bias (0.012 m^3/m^3), indicating a significantly smaller spread or variability of the estimation errors for the calibrated method. These results confirmed the feasibility of enhancing trapezoid-based SSM retrieval by calibrating the soil hydraulic characteristics over the trapezoid boundaries.

IV. DISCUSSION

A. On the Critical Value

The present study has preliminarily determined a critical value when comparing the PTF-derived field capacity and that obtained from long-term ground in situ SSM measurements over the REMEDHUS sites, as shown in Fig. 5. The importance of the critical value lies in its potential for dividing and correcting SSM overestimation and underestimation when the SSM was lower or higher, respectively. As seen from the results, it is clear that the critical (or threshold) value is generally determined from all the measurements over the REMEDHUS sites. Namely, it is more likely that the critical (or threshold) value has the potential to reflect local soil heterogeneity, rather than to describe any soil physical characteristic for a single soil. Hence, no relations between the threshold (or critical) value and physical characteristics for any REMEDHUS soil sample have been found thus far. Nevertheless, it is still expected that further evidence from more networks can provide useful information on the critical (or threshold) value.

Because the local soil texture measurements over the REMEDHUS region are obtained based on laboratory analyses of relatively high credibility, it is obvious that the accuracy of PTF-derived field capacity is most likely to play a decisive role on the determination of the critical value. Hence, more

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efforts should be focused on the developments of more reliable PTF for generating soil hydraulic characteristics (especially for field capacity) from ground soil attributes, which has also been emphasized in the previous section. Besides, it should also be noted that high heterogeneity of local soil texture would increase the difficulties for the accurate determination of actual soil hydraulic characteristics since most of the current available remote sensing missions cannot well capture high spatial resolution soil attributes, especially over large regions. Although the major scientific finding of the present study is the necessity of calibrating PTF-derived field capacity and wilting point for enhancing SSM estimation, which can be fully illustrated following the results for scheme (a), another issue with respect to the critical (or threshold) value can be the potential advantages of scheme (b) when SSM is with a high level (e.g., after heavy rain or intensive irrigation) or the study region is under humid climate zone. Following the results presented in Fig. 7, scheme (b) is most likely to reveal better SSM estimates than scheme (a) when SSM exceeds the critical (or threshold) value, although the scatterplots over the critical (or threshold) value take only a minority proportion. This is generally because the REMEDHUS region is characterized as a typical semiarid climate pattern with relatively low annual precipitation ($\sim$385 mm). Nevertheless, scheme (b) can be a potential alternative when the study region is with the opposite situation, although more evidence is required in future investigations.

Because the long-term ground in situ SSM measurements is not always available in practice, it is still interesting to develop more practical approaches for the determination of the critical value and subsequently to correct SSM retrieval with the trapezoid-based methods. To this end, it is encouraged to carry out more synchronous field experiments under different soil and water conditions with the unmanned aerial vehicle observations, so as to find better solutions for linking the soil hydraulic characteristics and remote sensing measurements at pixel scale [42], [43].

B. Calibration of PTF

In the satellite trapezoid-based SSM retrieval methods, the SSM status over the trapezoid dry and trapezoid wet edge is commonly regarded as the wilting point and the field capacity, respectively. Moreover, the wilting point and field capacity are generally determined by PTF in practice since many soil surveys or digital soil texture data are available nowadays, allowing for an unprecedented opportunity to obtain soil hydraulic parameters in regional or global scale with multiple spatial scales. Currently, a number of PTF has been developed by soil science community with each own advantages and drawbacks, making it difficult to determine a most adequate one for a specific region [27]. To this end, it is crucial to calibrate PTF-derived soil hydraulic parameters when applying them at satellite pixel scale, due to the spatial heterogeneity of soils and the inconsistency of various PTF.

The availability of local soil texture measurements and long-term in situ SSM observations for each REMEDHUS site provides an adequate opportunity to calibrate PTF-derived soil hydraulic characteristics (field capacity and wilting point) over the study region. With the calibration of the PTF-derived wilting point for the trapezoid dry edge and the PTF-derived field capacity for the trapezoid wet edge, enhanced SSM estimates were obtained using the trapezoid-based methods. Except for the calibration of the PTF-derived soil hydraulic parameters, another alternative can be the calibration of the PTF itself or developing new PTF. However, it is very complicated to calibrate PTF itself or to develop a new PTF in the present study because much more factors, rather than merely the soil texture and SSM measurements, should be acquired in advance. Moreover, to calibrate PTF itself or to develop new PTF, a large amount of data records are commonly necessary because of the requirements to further reflect the true population of soils in a region or the world. Hence, the present study mainly focused on the calibration of the PTF-derived soil hydraulic characteristics using local measurements. This is practical over the study region. Nevertheless, it is still expected that more universal PTF would be developed for applications in Earth system science in future developments.

Another notable issue in the calibration process is the use of long-term (2005–2021) SSM measurements for the calibration of the PTF-derived hydraulic characteristics, whereas the calibrated results were tested over a study period from 2013 to 2015. Although hydraulic characteristics are independent of SSM measurements in theory, it is still interesting to conform to the general principle of cross validation, in which the calibration period and validation period should be strictly separated. Fig. 9 shows the comparison of the maximum and minimum SSM values derived from two calibration periods from 2005 to 2021 and from 2005 to 2012, respectively. Except for the considerable correlation of 0.960 and 0.995, no significant differences for the two datasets can be found. The overall RMSE (bias) for the maximum SSM and minimum SSM is 0.007 m^3/m^3(-0.002 m^3/m^3) and 0.005 m^3/m^3(-0.003 m^3/m^3), respectively. These results indicate that for most of the REMEDHUS sites, it not always requires long-term in situ SSM measurements for the calibration of the PTF-derived soil hydraulic characteristics. This is generally reasonable because regular precipitation and/or irrigation can commonly lead to considerable SSM variation between field capacity and wilting point, which can be observed by SSM sensors. Nevertheless, for the purpose of calibrating PTF-derived soil hydraulic characteristics with continuous ground in situ SSM measurements, it is still recommended that field observations should cover at least an entire dry season and a wet season.

Fig. 10 shows the scatterplots between estimated SSM and in situ measurements over the study period from 2013 to 2015 over the REMEDHUS region, in which the PTF-derived soil hydraulic characteristics were calibrated with in situ SSM during the period from 2005 to 2012. Compared to those calibrated with in situ SSM measurements over the period from 2005 to 2021 shown in Fig. 8, it is obvious that similar error statistics with RMSE of 0.050 m^3/m^3, bias of $\sim$0.010 m^3/m^3, and R of 0.72 can be obtained. The results

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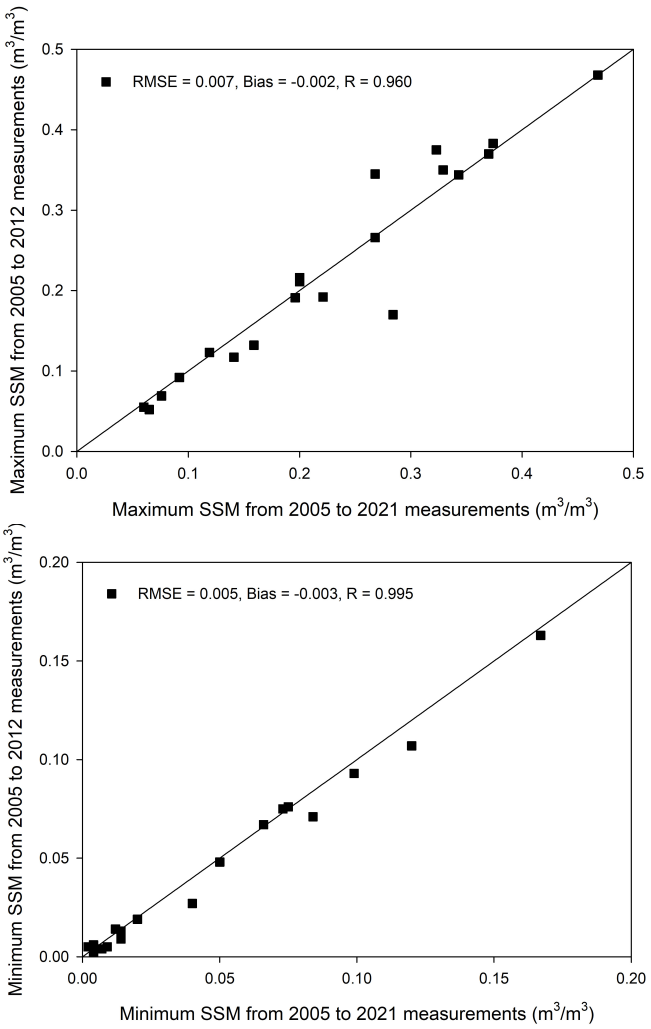


Fig. 9. Comparison of the maximum and minimum SSM values derived from different periods of ground in situ measurements.

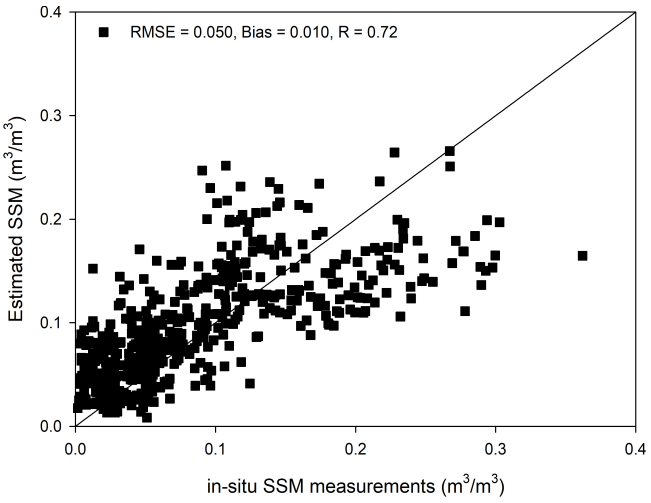


Fig. 10. Comparison of estimated SSM and in situ measurements. The PTF-derived soil hydraulic characteristics by Saxton et al. [29] were calibrated with in situ SSM measurements (2005–2012).

have further confirmed the feasibility of the proposed approach for improving SSM estimates with separated calibration period from validation period.

V. Conclusion

This study investigated SSM retrieval using the trapezoid method and provided a practical approach to calibrate PTF to obtain better soil hydraulic characteristics for SSM status determination over trapezoidal dry and wet edges. Compared to the uncalibrated PTF with an RMSE of 0.062 m^3/m^3 and a bias of 0.040 m^3/m^3 for the trapezoid-based SSM retrievals, the integration of the calibrated PTF and trapezoid method revealed an enhanced SSM retrieval with an RMSE of 0.050 m^3/m^3 and a bias of 0.012 m^3/m^3. Moreover, the phenomenon of overestimation and underestimation when the SSM was lower or higher than a certain threshold value, respectively, was also significantly improved. This is especially important in semiarid regions, where accurate SSM information is crucial for maintaining ecosystem services and benefiting agricultural activities. Finally, regarding the spatial extrapolation of PTF-based soil hydraulic characteristics into remote sensing models and other soil/crop models and the lack of ground measurements for local calibration over data-scarce regions, efforts are encouraged to focus on the development of more spatially representative PTF for describing soil water retention characteristics.

References

[1] S. I. Seneviratne et al., "Investigating soil moisture–climate interactions in a changing climate: A review," *Earth-Sci. Rev.*, vol. 99, pp. 125–161, May 2010.

[2] S. Zhou et al., "Soil moisture–atmosphere feedbacks mitigate declining water availability in drylands," *Nature Climate Change*, vol. 11, pp. 38–44, Jan. 2021.

[3] C. Champagne, J. White, A. Berg, S. Belair, and M. Carrera, "Impact of soil moisture data characteristics on the sensitivity to crop yields under drought and excess moisture conditions," *Remote Sens.*, vol. 11, no. 4, p. 372, Feb. 2019.

[4] J. M. Domínguez-Niño, J. Oliver-Manera, J. Girona, and J. Casadesús, "Differential irrigation scheduling by an automated algorithm of water balance tuned by capacitance-type soil moisture sensors," *Agricult. Water Manage.*, vol. 228, Feb. 2020, Art. no. 105880.

[5] P. D'Odorico, K. Caylor, G. S. Okin, and T. M. Scanlon, "On soil moisture-vegetation feedbacks and their possible effects on the dynamics of dryland ecosystems," *J. Geophys. Res., Biogeosci.*, vol. 112, no. G4, Dec. 2007, Art. no. G04010.

[6] L. Liu, L. Gudmundsson, M. Hauser, D. Qin, S. Li, and S. I. Seneviratne, "Soil moisture dominates dryness stress on ecosystem production globally," *Nature Commun.*, vol. 11, no. 1, p. 4892, Dec. 2020.

[7] Q. Zhang, W. Wei, L. Chen, L. Yang, Y. Luo, and A. Cai, "Plant traits in influencing soil moisture in semiarid grasslands of the loess plateau, China," *Sci. Total Environ.*, vol. 718, May 2020, Art. no. 137355.

[8] Z.-L. Li, P. Leng, C. Zhou, K.-S. Chen, F.-C. Zhou, and G.-F. Shang, "Soil moisture retrieval from remote sensing measurements: Current knowledge and directions for the future," *Earth-Sci. Rev.*, vol. 218, Jul. 2021, Art. no. 103673.

[9] S. H. Alemohammad, J. Kolassa, C. Prigent, F. Aires, and P. Gentine, "Global downscaling of remotely sensed soil moisture using neural networks," *Hydrol. Earth Syst. Sci.*, vol. 22, no. 10, pp. 5341–5356, 2018.

[10] B. Fang, V. Lakshmi, M. Cosh, P. Liu, R. Bindlish, and T. Jackson, "A global 1-km downscaled SMAP soil moisture product based on thermal inertia theory," *Vadose Zone J.*, vol. 21, no. 2, 2022, Art. no. e20182.

[11] J. Peng, A. Loew, O. Merlin, and N. Verhoest, "A review of spatial downscaling of satellite remotely sensed soil moisture," *Rev. Geophys.*, vol. 55, no. 2, pp. 341–366, 2017.

[12] N. Das et al., "The SMAP and Copernicus Sentinel 1A/B microwave active-passive high resolution surface soil moisture product," *Remote Sens. Environ.*, vol. 233, Art. no. 111380, Nov. 2019.

[13] B. Bauer-Marschallinger et al., "Soil moisture from fusion of scatterometer and SAR: Closing the scale gap with temporal filtering," *Remote Sens.*, vol. 10, no. 7, p. 1030, 2018.

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[14] B. Bauer-Marschallinger et al., "Toward global soil moisture monitoring with Sentinel-1: Harnessing assets and overcoming obstacles," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 520–539, Jan. 2019.
[15] G. Petropoulos, T. N. Carlson, M. J. Wooster, and S. Islam, "A review of Ts/VI remote sensing based methods for the retrieval of land surface energy fluxes and soil surface moisture," *Prog. Phys. Geogr., Earth Environ.*, vol. 33, no. 2, pp. 224–250, 2009.
[16] E. Babaeian, M. Sadeghi, S. B. Jones, C. Montzka, H. Vereecken, and M. Tuller, "Ground, proximal, and satellite remote sensing of soil moisture," *Rev. Geophys.*, vol. 57, no. 2, pp. 530–616, 2019.
[17] W. Zhao, N. Sanchez, and A. Li, "Triangle space-based surface soil moisture estimation by the synergistic use of in situ measurements and optical/thermal infrared remote sensing: An alternative to conventional validations," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 8, pp. 4546–4558, Aug. 2018.
[18] K. S. Rawat, V. K. Sehgal, S. K. Singh, and S. S. Ray, "Soil moisture estimation using triangular method at higher resolution from MODIS products," *Phys. Chem. Earth, A/B/C*, vol. 126, Jun. 2022, Art. no. 103051.
[19] W. W. Verstraeten, F. Veroustraete, C. J. van der Sande, I. Grootaers, and J. Feyen, "Soil moisture retrieval using thermal inertia, determined with visible and thermal spaceborne data, validated for European forests," *Remote Sens. Environ.*, vol. 101, no. 3, pp. 299–314, Apr. 2006.
[20] L. Fan et al., "Evaluation of the airborne CASI/TASI Ts-VI space method for estimating near-surface soil moisture," *Remote Sens.*, vol. 7, no. 3, pp. 3114–3137, 2015.
[21] P. Leng, Z.-L. Li, S.-B. Duan, M.-F. Gao, and H.-Y. Huo, "A practical approach for deriving all-weather soil moisture content using combined satellite and meteorological data," *ISPRS J. Photogramm. Remote Sens.*, vol. 131, pp. 40–51, Sep. 2017.
[22] A. J. Gijsman, S. S. Jagtap, and J. W. Jones, "Wading through a swamp of complete confusion: How to choose a method for estimating soil water retention parameters for crop models," *Eur. J. Agronomy*, vol. 18, nos. 1–2, pp. 77–106, Dec. 2002.
[23] H. Vereecken et al., "Modeling soil processes: Review, key challenges, and new perspectives," *Vadose Zone J.*, vol. 15, no. 5, pp. 1–57, 2016.
[24] J. A. Vrugt, P. H. Stauffer, T. Wöhling, B. A. Robinson, and V. V. Vesselinov, "Inverse modeling of subsurface flow and transport properties: A review with new developments," *Vadose Zone J.*, vol. 7, no. 2, pp. 843–864, May 2008.
[25] Y. Zha et al., "A reduced-order successive linear estimator for geostatistical inversion and its application in hydraulic tomography," *Water Resour. Res.*, vol. 54, pp. 1616–1632, Mar. 2018.
[26] J. Bouma, *Using Soil Survey Data for Quantitative Land Evaluation*, B. A. Stewart, Ed. New York, NY, USA: Springer, 1989, pp. 177–213.
[27] L. K. Van et al., "Pedotransfer functions in Earth system science: Challenges and perspectives," *Rev. Geophys.*, vol. 55, pp. 1199–1256, Dec. 2017.
[28] Y. Zhang and M. Y. Schaap Zha, "A high-resolution global map of soil hydraulic properties produced by a hierarchical parameterization of a physically based water retention model," *Water Resour. Res.*, vol. 54, pp. 9774–9790, Dec. 2018.
[29] K. Saxton, W. Rawls, J. Romberger, and R. Papendick, "Estimating generalized soil-water characteristics from texture," *Soil Sci. Soc. Amer. J.*, vol. 50, pp. 1031–1036, Jul./Aug. 1986.
[30] K. E. Saxton and W. J. Rawls, "Soil water characteristic estimates by texture and organic matter for hydrologic solutions," *Soil Sci. Soc. Amer. J.*, vol. 70, no. 5, pp. 1569–1578, Sep. 2006.
[31] P. J. Shellito, E. E. Small, and M. H. Cosh, "Calibration of Noah soil hydraulic property parameters using surface soil moisture from SMOS and basinwide in situ observations," *J. Hydrometeorol.*, vol. 17, no. 8, pp. 2275–2292, Aug. 2016.
[32] W. Sun, X. Yao, N. Cao, Z. Xu, and J. Yu, "Integration of soil hydraulic characteristics derived from pedotransfer functions into hydrological models: Evaluation of its effects on simulation uncertainty," *Hydrol. Res.*, vol. 47, no. 5, pp. 964–978, Oct. 2016.
[33] C. Sung and J. Iba, "Accuracy of the Saxton-Rawls method for estimating the soil water characteristics for mineral soils of Malaysia," *Pertanika J. Tropical Agricult. Sci.*, vol. 33, no. 2, pp. 297–302, 2010.
[34] Á. González-Zamora, N. Sánchez, M. Pablos, and J. Martínez-Fernández, "CCI soil moisture assessment with SMOS soil moisture and in situ data under different environmental conditions and spatial scales in Spain," *Remote Sens. Environ.*, vol. 225, pp. 469–482, May 2019.
[35] W. Dorigo et al., "Evaluation of the ESA CCI soil moisture product using ground-based observations," *Remote Sens. Environ.*, vol. 162, pp. 380–395, Jun. 2015.
[36] M. Cook, J. R. Schott, J. Mandel, and N. Raqueno, "Development of an operational calibration methodology for the Landsat thermal data archive and initial testing of the atmospheric compensation component of a land surface temperature (LST) product from the archive," *Remote Sens.*, vol. 6, no. 11, pp. 11244–11266, 2014.
[37] N. K. Malakar, G. C. Hulley, S. J. Hook, K. Laraby, M. Cook, and J. R. Schott, "An operational land surface temperature product for Landsat thermal data: Methodology and validation," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 10, pp. 5717–5735, Oct. 2018.
[38] L. Prihodko and S. N. Goward, "Estimation of air temperature from remotely sensed surface observations," *Remote Sens. Environ.*, vol. 60, no. 3, pp. 335–346, 1997.
[39] R. Tang, Z.-L. Li, and B. Tang, "An application of the Ts-VI triangle method with enhanced edges determination for evapotranspiration estimation from MODIS data in arid and semi-arid regions: Implementation and validation," *Remote Sens. Environ.*, vol. 114, pp. 540–551, Aug. 2010.
[40] M. Amani, B. Salehi, S. Mahdavi, A. Masjedi, and S. Dehnavi, "Temperature-vegetation-soil moisture dryness index (TVMDI)," *Remote Sens. Environ.*, vol. 197, pp. 1–14, Aug. 2017.
[41] A. Bablet et al., "MARMIT: A multilayer radiative transfer model of soil reflectance to estimate surface soil moisture content in the solar domain (400–2500 nm)," *Remote Sens. Environ.*, vol. 217, pp. 1–17, Nov. 2018.
[42] Y. Liu et al., "Soil texture and its relationship with environmental factors on the Qinghai–Tibet plateau," *Remote Sens.*, vol. 14, no. 15, p. 3797, 2022.
[43] Y. Zhai, J. A. Thomasson, J. E. Boggess, and R. Sui, "Soil texture classification with artificial neural networks operating on remote sensing data," *Comput. Electron. Agricult.*, vol. 54, no. 2, pp. 53–68, Dec. 2006.



Pei Leng (Senior Member, IEEE) received the Ph.D. degree in cartography and geographical information system from the University of Chinese Academy of Sciences, Beijing, China, in 2015.

He is currently an Associate Professor with the Key Laboratory of Agricultural Remote Sensing, Ministry of Agriculture and Rural Affairs/Institute of Agricultural Resources and Regional Planning, Chinese Academy of Agricultural Sciences, Beijing. His research interests include the methodological developments of satellite soil moisture and energy fluxes, and agricultural remote sensing.

Zhao-Liang Li (Fellow, IEEE) received the Ph.D. degree in geophysics and remote sensing from the University of Strasbourg, Strasbourg, France, in 1990.

He joined the Institute of Agricultural Resources and Regional Planning, Beijing, China, in 2013. Since 1992, he has been a Research Scientist with CNRS, Illkirch-Graffenstaden, France. He has participated in many national and international projects, such as NASA-funded MODIS, EC-funded program EAGLE, and ESA-funded program SPECTRA. His research interests include thermal infrared radiometry, parameterization of land surface processes at large scale, and the assimilation of satellite data to land surface models.

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Qian-Yu Liao is currently pursuing the Ph.D. degree in cartography and geographic information systems with the Institute of Geographic Sciences and Natural Resources Research, Chinese Academy of Sciences (CAS), Beijing, China.

His research interests include the retrieval of evapotranspiration and soil moisture from satellite observations.



Yun-Jing Geng received the B.E. degree in remote sensing science and technology from Shandong Normal University, Jinan, China, in 2020. She is currently pursuing the M.D. degree in agricultural remote sensing with the Institute of Agricultural Resources and Regional Planning, Chinese Academy of Agricultural Sciences, Beijing, China.

Her research interests include the retrieval of soil moisture and evapotranspiration from satellite observations.



Qiu-Yu Yan received the B.E. degree in land resources management and the M.E. degree in geological engineering from Hebei GEO University, Shijiazhuang, China, in 2019 and 2022, respectively.

She is currently with the Key Laboratory of Agricultural Remote Sensing, Ministry of Agriculture and Rural Affairs/Institute of Agricultural Resources and Regional Planning, Chinese Academy of Agricultural Sciences, Beijing, China. Her research interests include retrieval of soil moisture from satellite data and agricultural remote sensing.



Xia Zhang was born in Shijiazhuang, Hebei, China, in 1985. She received the B.S. degree in geographic information system from Changchun Normal University, Changchun, China, in 2008, and the M.S. and Ph.D. degrees in cartography and geographic information system from Northeast Normal University, Changchun, in 2010 and 2014, respectively.

Since 2014, she has been a Lecturer with the College of Land Resources and Urban and Rural Planning, Hebei GEO University, Shijiazhuang. Her research interests include thermal infrared polarization remote sensing, multiangle and hyperspectral remote sensing, and quantitative remote sensing.



Guo-Fei Shang was born in Tangshan, Hebei, China, in 1964. He received the B.E. degree in aerial photogrammetry and remote sensing from Wuhan University, Wuhan, China, in 1985, and the M.E. and Ph.D. degrees in land resources management from the Renmin University of China, Beijing, China, in 1992 and 2002, respectively.

Since 1997, he has been a Professor and the Dean of the College of Land Resources and Urban and Rural Planning, Hebei GEO University, Shijiazhuang, China. His research interests include aerial photogrammetry and remote sensing, quantitative remote sensing, and natural resources management.

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